# EV-220 Freezer Evaporator: Service Manual

| Document | Revision | Effective | Owner |
|---|---|---|---|
| NG-MAN-EV220 | C | 2025-03-01 | Northgate Equipment Co. |

## 1. Overview

The Northgate EV-220 is a low-profile electric-defrost evaporator for walk-in and warehouse freezers. It holds box temperatures between -15F and -5F. Defrost runs on a timer, 4 times per day by default, and ends on coil temperature (50F) or after a 45 minute fail-safe.

## 2. Specifications

| Item | Value |
|---|---|
| Capacity | 22,000 BTU/h at -20F SST, 10F TD |
| Fans | 3 x 16 in ECM motors, 230V / 1 ph |
| Defrost heaters | 4.8 kW total, electric |
| Drain line heater | Self-regulating, 8 W/ft |
| Power draw, normal | 1.5 to 2.2 kW (fans and controls); 6.5 kW during defrost |

## 3. Normal Operating Ranges

| Measurement | Normal range | Alarm limit |
|---|---|---|
| Box temperature | -15F to -5F | Above 0F for 30 min (F-20) |
| Suction pressure | 15 to 25 psig | Below 10 psig |
| Defrost duration | 18 to 30 min | Over 45 min (DF-4) |
| Fan vibration | Below 2.0 mm/s | Above 4.0 mm/s |

A box temperature that climbs during the night defrost window and keeps climbing after the window ends usually means a defrost that did not terminate. Check for DF-4.

## Fault Code Reference

| Code | Name | Trigger condition | Severity |
|---|---|---|---|
| DF-4 | Defrost termination failure | Defrost runs past the 45 minute fail-safe, or the coil sensor never reaches 50F. | High |
| F-11 | Evaporator fan failure | One or more fans draw under 0.2 A while commanded on. | Medium |
| F-20 | High box temperature | Box temperature above 0F for 30 consecutive minutes. | High |
| F-31 | Drain line heater fault | Drain heater current below 0.1 A. | Low |

## Troubleshooting by Fault Code

### *DF-4: Defrost termination failure*

**Trigger:** Defrost runs past the 45 minute fail-safe, or the coil sensor never reaches 50F.

**Likely causes, most common first:**

- Failed defrost termination sensor (most common).
- Defrost contactor stuck closed.
- Controller clock or schedule fault after a power loss.

**Corrective actions:**

1. Check the controller display. If it shows DEFROST more than 45 minutes after the start, force the unit back to cooling.
2. Measure the termination sensor resistance. It should read 10k ohm at 77F. Replace if open or shorted.
3. Check the defrost contactor. If the contacts are welded, replace the contactor.
4. Watch the next scheduled defrost to confirm it ends between 18 and 30 minutes.
5. If box temperature went above 0F, start SOP-OPS-002.

Heaters left on during a failed defrost can raise box temperature by 20F or more in 3 hours.

### *F-11: Evaporator fan failure*

**Trigger:** One or more fans draw under 0.2 A while commanded on.

**Likely causes, most common first:**

- Failed ECM motor.
- Ice on fan blade.
- Loose plug at motor harness.

**Corrective actions:**

1. Look for ice on the fan guard and run a manual defrost.
2. Check the harness plug.
3. Replace the motor if it does not start with 230V present.

### *F-20: High box temperature*

**Trigger:** Box temperature above 0F for 30 consecutive minutes.

**Likely causes, most common first:**

- Active DF-4 or F-11.
- Door held open during loading.
- Iced coil.

**Corrective actions:**

1. Resolve any other active fault.
2. Close doors and check strip curtains.
3. Start SOP-OPS-002. Freezer product above 10F for more than 60 minutes must be quarantined.

### *F-31: Drain line heater fault*

**Trigger:** Drain heater current below 0.1 A.

**Likely causes, most common first:**

- Open heater cable.
- Tripped GFCI.

**Corrective actions:**

1. Reset the GFCI.
2. Replace the heater cable if open. Check for ice in the drain pan.

## Preventive Maintenance

| Task | Interval |
|---|---|
| Inspect coil for ice pattern | Monthly |
| Test defrost termination sensor | Quarterly |
| Check drain line heater | Before winter, then quarterly |
| Clean fan blades and guards | Semi-annually |

**WARNING:** Apply lockout/tagout per SOP-OPS-001 before opening any electrical enclosure or touching moving parts. Only authorized technicians may perform these steps.